

DRP Final Writeup

My DRP focused on Decision Theory and was mentored by Jasper Yang. We spent the first half of the quarter reading and discussing *Decision Theory: Principles and Approaches* by Parmigiani and Inoue. This served as my introduction to a new scope of statistics, namely the divide between frequentist and Bayesian perspectives, utility in practice, and the discussion of numerous decision problems in a statistical context.

Topics we discussed included rationality and coherence assumptions for decision theory, expected utility theorem, loss and risk functions, and Allais and Ellsberg's paradoxes. We then discussed scoring rules, measures by which to assess and 'predictors and decision making frameworks.

For the second half of the quarter, I focused on researching the concepts of the 11th chapter further: how do we choose models? The chapter raised questions about how we choose distributions in our probability questions, and whether or not choosing just one distribution or model was correct at all. The book discussed alternative methods like Bayesian Model Averaging (under the true model perspective) or Model Elaboration (under the open model perspective). I did further research into Bayesian Model Averaging by reading *Bayesian Model Averaging: A Tutorial* by Hoeting et al, which became an element of my presentation.

The DRP concluded with Höeppner's 2026 work on modeling decision making under uncertainty, which discussed the dominant decision making frameworks, their strengths and weaknesses, and their best applications. Frameworks included the modern expected utility theorem, heuristics, risk sensitivity, and fuzzy set theory. The ultimate conclusion was that, whether it is choosing an action or choosing the framework with which to choose actions, decision theory problems arise everywhere in life.

Though there was much more to learn from my first exposure to research than I expected, it was a valuable experience.